Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Next-Generation Drug Target Discovery Platform Integrating Cell-Type-Specific and Spatial Proteomics with AI			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Pharmaceutical Bio-based Technologies and Systems	Bio-processes / Bio-Equipment	Bio-process Technologies
Concept of Proposed Technology	The proposed technology aims to establish a comprehensive drug target discovery platform that integrates cell-type-specific[1] and spatial proteomics[2] with AI-based data interpretation to enable precise, context-aware identification of actionable drug targets. Recent advances in cell-type-specific protein labeling and high-resolution data-independent acquisition[3] (DIA) mass spectrometry enable selective enrichment and quantification of proteins originating from specific cellular populations within complex tissues. These experimental workflows generate large-scale, biologically rich datasets that capture cellular identity and microenvironmental context, offering substantial potential for drug discovery. However, such datasets are inherently complex, characterized by partial enrichment, background contamination, peptide interference, and context-dependent expression patterns, which exceed the analytical capabilities of conventional proteomics softwares[4,5] optimized for bulk samples. As a result, there exists a critical gap between experimental innovation and practical drug discovery output.			

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	To address this challenge, the proposed project integrates advanced proteomics workflows with a proprietary AI-based DIA analysis framework capable of cell-type-specific, and spatially informed data interpretation. The integrated platform explicitly models enrichment efficiency, background signals, and cellular specificity, transforming complex proteomics data into quantitative, metrics that directly support drug target identification, mechanism-of-action (MoA) analysis, and early off-target risk assessment. Integrating experimental proteomics and AI-driven analytics, the resulting platform is designed to be scalable, reproducible, and adaptable pipeline, serving as a foundation for next-generation, cell-resolved drug discovery across diverse disease areas.
Rationale for Joint R&D Project	The research institute possesses advanced capabilities in cell-type-specific proteomics, spatial tissue analysis, and disease-relevant experimental models, enabling the generation of high-quality, biologically rich proteomics datasets. However, the scale and complexity of these datasets create a data interpretation bottleneck that cannot be efficiently addressed using conventional academic analysis workflows. Conversely, the Korean SME partner is expected to possess a proprietary AI-based DIA proteomics analysis platform with demonstrated capacity to process large-scale proteomics data for drug discovery or chemoproteomics applications. This capability enables rapid development of analytical frameworks that incorporate enrichment-aware logic, cell-type specificity, and spatial context. Joint R&D is essential to bridge experimental and computational domains through iterative co-development, where experimental insights inform algorithmic refinement and AI outputs guide subsequent biological validation. This collaboration supports the creation of a globally competitive drug target discovery platform aligned with national priorities in advanced bio–AI convergence while strengthening the domestic SME ecosystem.
Key Words	1. Cell-type-specific proteomics 2. Data-independent acquisition (DIA) analysis 3. AI-based protein identification and quantification 4. Drug target discovery 5. Chemoproteomics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	• Establish an integrated Bio–AI workflow linking cell-type-specific and spatial proteomics with AI-driven analytical pipelines • Optimize AI architectures for precision DIA processing, including enrichment-aware and library-free analysis • Enable multi-scale disease mechanism elucidation and identification of novel therapeutic targets • Improve early-stage assessment of off-target interactions and MoA profiles			

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R&D Target Specifications	**Analytical Performance** • Identification of >20% more proteins from standard DIA datasets compared with industry-standard softwares[3,4] • Enhanced detection and characterization of low-abundance proteins and post-translational modifications **Quantitative Accuracy** • Protein-level quantification coefficient of variation (CV) <15% across biological and technical replicates • Reliable detection of subtle expression changes (<0.5-fold differences) **Processing Throughput** • Reduction of computational processing time by >30% through AI-accelerated frameworks • Capability to process terabyte-scale MS datasets within high-throughput timelines
R&D Plan	**Year 1 – Experimental and Computational Integration** • Standardize cell-type-specific labeling and high-resolution DIA MS analysis workflows • Establish and validate independently developed DIA analysis capabilities suitable for enriched and heterogeneous proteomics datasets • Complete proof-of-concept integrated experimental–AI analysis runs **Year 2 – AI Model Optimization and Iterative Validation** • Train and refine deep learning models for peptide identification, fragmentation prediction, and retention time estimation • Establish secure AI software environments for large-scale MS data ingestion • Perform iterative feedback cycles between AI predictions and experimental validation **Year 3 – Platform Validation and Translational Application** • Validate platform performance using complex disease-relevant samples • Demonstrate identification of actionable drug targets and MoA insights • Prepare the platform for industrial deployment and global service offerings

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Generate high-quality cell-type-specific and spatial proteomics datasets • Lead biological interpretation and translational validation of identified targets • Provide technical feedback and domain expertise to refine AI models • Validate biological relevance of AI-derived insights through experimental assays

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Expected R&R of Korean SME Partner	• Develop and operate AI-based DIA proteomics analysis pipeline • Advance protein identification and quantification algorithms using AI • Integrate quality control, visualization, and reporting systems • Manage commercialization strategies and global intellectual property

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Impact** • Establishment of a world-class AI–proteomics drug discovery platform • Resolution of long-standing signal-to-noise challenges in DIA proteomics • Transition from descriptive proteomics to predictive, functional target discovery **Economic and Industrial Impact** • Reduction of dependency on foreign proteomics software platforms • Creation of exportable, high-value AI-based analysis services • Strengthening of domestic SME competitiveness in the global bio–AI market **Strategic and Societal Impact** • Acceleration of drug development through improved target deconvolution • Support for personalized and precision medicine initiatives • Alignment with national and global priorities in advanced bio and artificial intelligence convergence

6. References

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(3) Doerr, A. DIA mass spectrometry. Nat Methods 12, 35–35 (2015)

(4) Staes, A. et al. Benchmarking DIA data analysis workflows. bioRxiv 2023, doi:10.1101/2023.06.02.543441 (2023)

(5) Lou, R. et al. Benchmarking commonly used software suites and analysis workflows for DIA proteomics and phosphoproteomics. Nat. Commun. 14, 94 (2023)